Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: finite RES coefficients are insufficient. In the unlocalized graded tensor ring, the kernel ideal cannot be generated using only finite RES[0] coefficients. This requires allowing positive-dimensional X in the presentation; it does not assert that every individual X is indispensable.

Let J_{fin} be the ideal generated by all the earlier translation differences with a finite coefficient $[F]_0$, allowing arbitrary Gamma arity. Work over $\mathbb{S} = \mathbb{C}((t))$, $\mathrm{val}(t) = 1$, and put $\delta = 1/5$. Consider

$$X = \{(x, y) : x \in \Bbbk \setminus \{0, 1\},\ y \in \delta^\sharp,\ y^5 = \mathrm{rv}(t)x(x-1)\}.$$

Here nonzero residue-field elements are identified with value-zero RV elements. Projection $f(x, y) = x$ has five-element fibres, so $(X, f) \in \mathrm{RES}[1]$; the source has the single valuation tuple $(0, 1/5)$. The map $c(x, y) = y$ witnesses $\delta \in \Lambda_X$. With $H_a = (a, \infty)$, the degree-two element

$$z_X = [X]_1 \otimes \big([H_0]_1 - [H_{1/5}]_1\big) \quad \text{belongs to} \quad \ker \Psi,$$

by $(x, y, u) \mapsto (x, y, yu)$, whose inverse divides by y.

Why it is outside the finite-coefficient ideal. Choose a fifth root β of $\mathrm{rv}(t)$ only to examine the residue geometry, and write $y = \beta z$. The normalized curve is $z^5 = x(x-1)$. It is irreducible, with smooth projective normalization of genus two: the degree-five map to the x-line is totally ramified at $0, 1, \infty$, and $2g - 2 = -10 + 12 = 2$. The cyclic descent action $z \mapsto \zeta z$ preserves its unique curve component setwise.

For a finite RES object F, let $n(F)$ count its $\mathbb{S}$-definable points. For a RES[1] object Y, let $b(Y)$ count its positive-genus geometric curve components preserved setwise by the entire cyclic descent action. This is an additive invariant within the degree-one category: constructible bijections induce equivariant birational maps on dense curve components. It satisfies

$$b(F \times Y) = n(F)b(Y), \qquad b(a^\sharp) = 0, \qquad b(X) = 1.$$

Finite components and genus-zero curves contribute zero. The fixed grading is retained throughout this construction.

Let $D = \mathbb{Z}[\eta, \varepsilon]/(\eta^2, \varepsilon^2)$, with both variables of degree one; $\eta\varepsilon \neq 0$. Map the RES ring to D by $[F]_0 \mapsto n(F)$, $[Y]_1 \mapsto b(Y)\eta$, and all degrees at least two to zero. Map the Gamma ring in degree zero by cardinality, in degree one by $[U]_1 \mapsto \nu(U)\varepsilon$, and in higher degrees by zero. Here ν is the earlier odd, integer-periodic endpoint invariant, with $\nu(H_0) = 0$, $\nu(H_{1/5}) = 1$, and ν zero on finite sets. These are ring homomorphisms, and they agree on the whole finite-Gamma coefficient ring: its degree-one lifts are rational value fibres, and both sides give zero. Hence they define a balanced tensor detector $\Phi : T \longrightarrow D$.

If a finite F admits a witness $F \longrightarrow \alpha^\sharp$ and $n(F) > 0$, evaluating it at a definable point forces $\alpha \in \mathbb{Z}$. Then ν is invariant under that translation. If $n(F) = 0$, the coefficient itself maps to zero. Higher Gamma arities also map to zero. Thus $\Phi(J_{\mathrm{fin}}) = 0$, whereas

$$\Phi(z_X) = \eta(0 - \varepsilon) = -\eta\varepsilon \neq 0. \qquad \boxed{z_X \in \ker \Psi \setminus J_{\mathrm{fin}}.}$$

This separates the ideals even after arbitrary ring multiplication and groupification. Finite coefficients placed in higher degrees add nothing: their relations are multiples of their degree-zero versions by the corresponding graded singleton.

The connected finite cover in this example therefore carries geometry that the finite-coefficient relations miss. The detailed descent, genus and detector checks are recorded in the finite-coefficient verification note. These are mathematical proofs, not complete Lean exports; no HK theorem is taken as a premise, and the approved working manuscript remains unchanged.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of RV[$*$] then it may also denote the object $(\boldsymbol{U}, 0)$ of μRV[$*$] with the constant volume form 0.